

PEM 2022 Year 11 Mathematics Standard Yearly Examination

SUGGESTED SOLUTIONS

Section 1 Multiple Choice Answer Key

Question	Answer	Outcomes Assessed	Targeted Performance Bands
1	C	<i>MS11 – 6</i>	2 – 3
2	D	<i>MS11 – 1</i>	2 – 3
3	A	<i>MS11 – 6</i>	3 – 4
4	D	<i>MS11 – 3</i>	2 – 3
5	D	<i>MS11 – 3</i>	4 – 5
6	B	<i>MS11 – 3</i>	4 – 6
7	B	<i>MS11 – 7</i>	3 – 4
8	D	<i>MS11 – 5</i>	4 – 5
9	C	<i>MS11 – 4</i>	3 – 4
10	A	<i>MS11 – 6</i>	3 – 5
11	D	<i>MS11 – 5</i>	4 – 5
12	A	<i>MS11 – 7</i>	3 – 5

Section II (Total 68 marks)

Question 13 (a) F1.1

Outcomes Assessed: MS11– 5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct value for n	2
Provides correct substitution into formula	1

Sample Answer:

$$525 = 2500 \times 3\% \times n$$

$$\therefore n = \frac{525}{75}$$

$$\therefore n = 7 \text{ years}$$

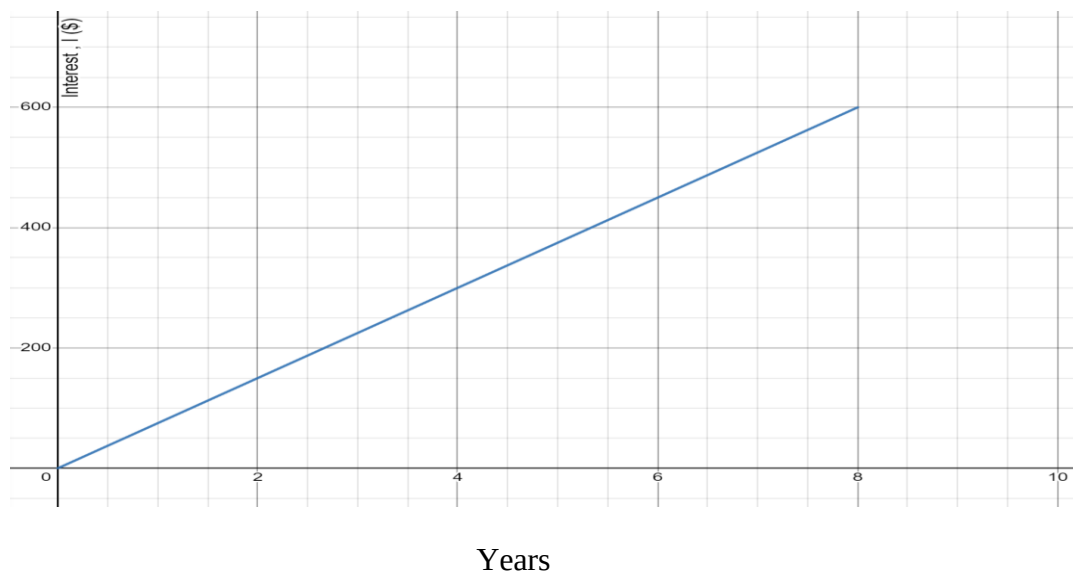
Question 13 (b) F1.1

Outcomes Assessed: MS11– 2

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct graph	2
Makes progress towards the solution by plotting 2 or more correct points	1

Sample Answer:



Question 14 (a) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	2
Make some progress towards using Pythagoras theorem	1

Sample Answer:

$$s^2 = 0.25^2 + 1.4^2$$

$$= 2.0225$$

$$s = 1.42 \text{ m}$$

Question 14 (b) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct answer	2
Makes progress towards answer	1

$$SA = 2 \times \frac{1}{2}h(a + b) + 2ls + al + bl$$

$$= 1.4(1.5 + 2) + 2 \times 3 \times 1.42 + 1.5 \times 3 + 2 \times 3$$

$$= 23.92 \text{ m}^2$$

Question 14 (c) M1.2

Outcomes Assessed: MS11– 3

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$23.92 \times 0.73 = \$ 17.46$$

Question 15 A1

Outcomes Assessed: MS11– 1

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct answer to 3 sig. figs.	1

Sample Answer:

$$\begin{aligned} (3.2 \times 10^9) \div (2.3 \times 10^7) &= 139.13047 \\ &= 139 \text{ (to 3 sig. figs.)} \end{aligned}$$

Question 16 (a) M1.3

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct answer	1

Sample Answer:

$$B = \frac{44.9}{1.56^2} = 18.45$$

Question 16 (b) M1.3

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct answer	1

Sample Answer : Yes

Question 17 (a) A2

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct gradient and y-intercept	2
Provides formula for gradient or other working	1

Sample Answer:

$$m = \frac{-100}{5} = -20$$

y – intercept 100

Question 17 (b) A2

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct temperature	1

Sample Answer: 40°C

Question 17 (c) A2

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct time	1

Sample Answer: 2.5 hours

Question 18 (a) S2

Outcomes Assessed: MS11– 7

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer: 90**Question 18 (b) S2**

Outcomes Assessed: MS11– 7

Targeted Performance Band: 3-4

Criteria	Marks
Correct answer	1

Sample Answer:

3 times a week

Question 18 (c) S2

Outcomes Assessed: MS11– 7

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards answer	1

Sample Answer:

$$\frac{18}{90} = 0.2$$

Question 19 (a) M1.2

Outcomes Assessed: MS11– 3

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution, rounded correctly	2
Makes progress towards the answer by demonstrating use of pythagoras	1

Sample Answer:

$$DC = \sqrt{39^2 + 15^2}$$

$$= 41.785 \dots$$

$$= 41.8 \text{ m}$$

Question 19 (b) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Provides some relevant calculation	1

Sample Answer:

$$A = \frac{1}{2} \times 94 \times 29 + \frac{1}{2} \times 15 \times 39 + \frac{1}{2} \times 32(39 + 35) + \frac{1}{2} \times 47 \times 35$$

$$= 3662 \text{ m}^2$$

Question 20 S2

Outcomes Assessed: MS11– 8

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution by finding difference in size or percent.	1

Sample Answer:

$$P(GGB) = \frac{10}{12} \times \frac{9}{11} \times \frac{2}{10} = \frac{3}{22}$$

$$P(GBG) = \frac{10}{12} \times \frac{2}{11} \times \frac{9}{10} = \frac{3}{22}$$

$$P(BGG) = \frac{2}{12} \times \frac{10}{11} \times \frac{9}{10} = \frac{3}{22}$$

$$Total = \frac{9}{22}$$

Question 21

Outcomes Assessed: MS11-3

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct quarterly bill	3
Calculate weekly cost	2
Calculates correct kWh	1

Sample Answer:

$$2800W = 2800 \times 0.001kW = 2.8kW$$

$$2.8kW \times 3.5h \times 6 = 58.8kWh$$

$$58.8 \times 13.189 = 775.51 \text{ cents} = \$7.75 \text{ per week}$$

$$7.76 \times 12 \text{ weeks} = \$93.12$$

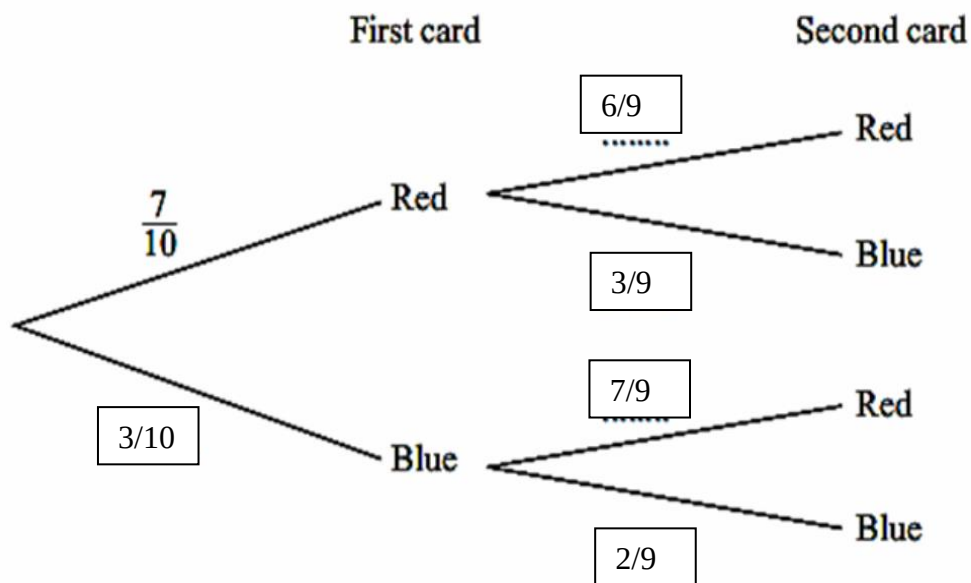
Question 22(a) M1.3

Outcomes Assessed: MS11-8

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct probabilities	3
Provides 4-5 correct probabilities	2
Provides 1 correct probability	1

Sample Answer:



Question 22 (b) M1.3

Outcomes Assessed: MS11-8

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards answer	1

Sample Answer:

$$\begin{aligned}P(RR) + P(BB) &= \frac{7}{10} \times \frac{6}{9} + \frac{3}{10} \times \frac{2}{9} \\&= \frac{42}{90} + \frac{6}{90} \\&= \frac{8}{15}\end{aligned}$$

Question 23 (a) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct answer	1

Sample Answer:

$$123409 \times 0.06 = \$ 7404.54$$

Question 23 (b)

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct answer	1

Sample Answer:

$$2.3\% \times 7404.54 = \$ 170.30$$

Question 23 (c)

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$\$ 7404.54 - \$ 170.30 = \$ 7234.24$$

$$\$ 7234.25 \div 52 = \$ 139.12$$

Question 24 (a) M2

Outcomes Assessed: MS11-3

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Provides one correct part	1

Sample Answer:

Katoomba, 15 hours ahead

Question 24 (b) M2

Outcomes Assessed: MS11-3

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct day and time	2
Makes progress towards answer	1

Sample Answer:

From the clocks, it will be midnight in New York in 2 hours 37 minutes and 17 seconds or 3 hours plus 12 hours forward.

So, time in Katoomba will be 3pm on Saturday

Question 25 (a) F1.2

Outcomes Assessed: MS11-3

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct answer	2
Makes progress towards answer	1

Sample Answer:

$$\text{\$ } 21.16 \times 20 \times 44 = \text{\$ } 18620.80$$

Question 25 (b) M2

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Calculates tax payable	2
Makes progress towards finding correct tax	1

Sample Answer:

$$\text{Taxable income} = 18620.80 - 250 = \text{\$}18\,370.80$$

$$18370.80 - 18200 = \text{\$}170.80$$

$$0.19 \times 170.80 = \text{\$}32.45$$

Question 25 (c) M2

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Answers question correctly	1

No

Question 26 (a) S1

Outcomes Assessed: MS11-7

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct mean	2
Makes progress toward answer	1

Sample Answer:

Length (mm)	Frequency	Midpoint x_i	$x_i f_i$
25-29	2	27	54
30-34	4	32	128
35-39	7	37	259
40-44	10	42	420
45-49	8	47	376
50-54	6	52	312
55-59	3	57	171
	40		1720

$$\text{Mean} = \frac{1720}{40} = 43$$

Question 26 (b)

Outcomes Assessed: MS11-7

Targeted Performance Band: 4-6

Criteria	Marks
Provides correct median	2
Find which score is the median	1

Sample Answer: 20th and 21st scores i.e. 40-44, 42

Question 26 (c)

Outcomes Assessed: MS11-7

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct class	1

Sample Answer:

Modal class is 40-44

Question 27 (a) S1

Outcomes Assessed: MS11-6

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Identifies correct percentage of population	1

Sample Answer:

Income from Government pensions is orange bar

Which is 23%

23% of 18230100 = 4192923

Question 27 (b) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Shows progress towards correct answer	1

Sample Answer:

$$\frac{18230100}{25727066} = 70.86\%$$

Question 28 (a) S1.1

Outcomes Assessed: MS11-2

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$90\% - 40\% = 50\%$$

Question 28 (b) S1.1

Outcomes Assessed: MS11-2

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct answer	1

Sample Answer:

Cuff defect

Question 29 (a) F1.1

Outcomes Assessed: MS11-5

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$10000 \times 0.05 = 500$$

Question 29 (b) F1.1

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct entries into table	2
Provides some correct entry into table	1

Sample Answer:

Year	Interest	Amount in account
0		\$10000
1	500	10500
2	525	11025
3	551.25	11576.25
4	578.81	12155.06
5	607.75	12762.81

Question 29 (c) F1.1

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Calculates correct difference	3
Find the total compounded monthly	2
Convert r and n to monthly	1

Sample Answer:

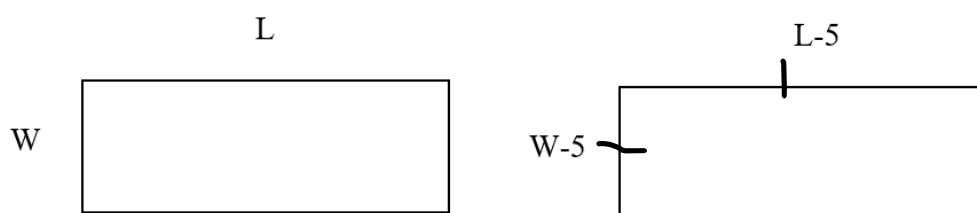
$$\begin{aligned}
 P &= A(1+r)^n, \quad r = 5\% \div 12 = 0.00417 \\
 &= 10000(1.00417)^{5 \times 12} \\
 &= 12833.59 \\
 \$12833.59 - \$12762.81 &= \$70.78
 \end{aligned}$$

Question 30 (a) A2

Outcomes Assessed: MS11-10

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct steps to form both equations	2
Shows one correct equation	1

Sample Answer:

$$\text{Perimeter} = 2L + 2W = 200\text{cm}$$

becomes a square so $L-5 = W-5$ **Question 30 (b) A2**

Outcomes Assessed: MS11-1

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

$$2L + 2W = 200 \quad (1)$$

$$L - 5 = W + 5 \rightarrow L = W + 10 \quad (2)$$

Substitute into (1)

$$2(W + 10) + 2(W) = 200$$

$$4W + 20 = 200$$

$$4W = 180$$

$$W = 45$$

Therefore, the side of the square will be $W + 5 = 50\text{cm}$.